

Relatório Técnico - Laboratório MPLS, VPLS e OSPF no MikroTik

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Mauricio
Junior

Requisitos:

- **Infraestrutura e Roteamento (Backbone):**

- Executar a rede utilizando roteadores MikroTik.
- Rede em MPLS.
- Aplicar de forma idêntica (simétrica) os custos definidos para as conexões em ambos os lados de cada enlace.

- **Serviço L2VPN (Túnel L2):**

- Estabelecer um túnel de Camada 2 (L2) entre o R8 e o R1.
- Conectar um equipamento em cada extremidade desse túnel (nos switches associados ao R1 e R8) e garantir a comunicação direta em L2 entre eles.

- **Servidor e Aplicações:**

- Adicionar um servidor Ubuntu à topologia.
- Deixar o servidor plenamente acessível pela rede e com o serviço web Nginx instalado e rodando.

- **Evidências e Documentação:**

- Capturar prints das adjacências estabelecidas em todos os equipamentos da rede.
- Executar e registrar testes de `ping` e `traceroute` entre o R3 e o R7 em ambas as direções (R3 -> R7 e R7 -> R3).

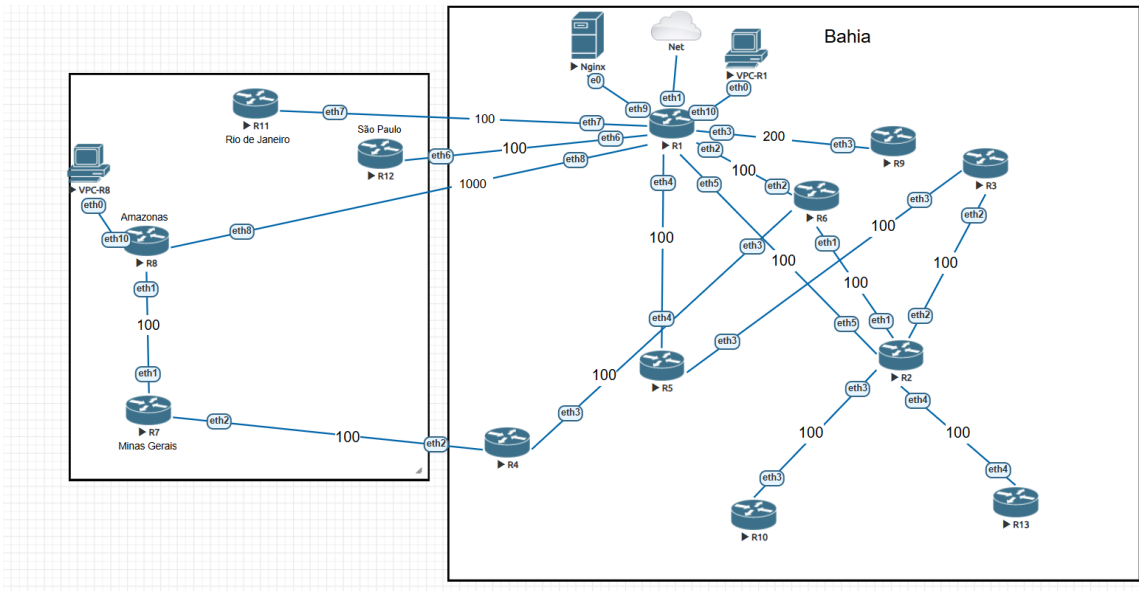
- **Gerenciamento:**

- Efetuar o backup completo das configurações de todos os roteadores para permitir restauração rápida caso necessário.

1.Topologia:

- **Visão Geral:** A rede é composta por 13 Routers Mikrotik compartilhando rotas via OSPF com métricas de custo e autenticação definidos. O MPLS está sendo usado para o encaminhamento de tráfego, também foi estabelecido uma VPLS entre R1 e R8 para que os equipamentos da Lan se comunicassem via Layer 2, atribuindo regras de firewall para que não houvesse comunicação da Lan com os Routers pelos endereços de Loopback ou pelos endereços estabelecidos nas interfaces físicas dos routers. Simulando uma aplicação, foi adicionado um servidor Linux com Nginx que está acessível por toda a rede, incluindo a Lan do R1 e do R8. Por fim, todos os dispositivos conseguem se comunicar com a internet.

- **Diagrama da Rede:**



2.Nomeclaturas:

• R1-13	• Roteadores da infraestrutura numerados de 1 a 13
• VPC-R1-R8	• Máquinas do túnel VPLS R1 e R8
• Net	• Conexão com a Internet
• Nginx	• Aplicação Web
• 100;200;1000	• Custos OSPF
• Estados	• Indica a região onde estão localizados os equipamentos

3.Ligação das Interfaces:

Para facilitar a gerência e a configuração dos equipamentos, a ligação física dos Routers seguiu o seguinte padrão: Ambas as extremidades utilizam a mesma numeração, criando uma simetria. R1 ether2 : R6 ether2

R1	R8		R2	R13
Ether8	Ether8		Ether4	Ether4
R1	R6		R2	R10
Ether2	Ether2		Ether3	Ether3
R1	R11		R2	R3
Ether7	Ether7		Ether2	Ether2
R1	R12		R2	R6
Ether6	Ether6		Ether1	Ether1
R1	R9		R6	R4
Ether3	Ether3		Ether3	Ether3
R1	R5		R5	R3
Ether4	Ether4		Ether3	Ether3
R1	R2		R7	R4
Ether5	Ether5		Ether2	Ether2
R8	R7		R1	Nginx
Ether1	Ether1		Ether9	Ether0
R1	VPC-R1		R8	VPC-R8
Ether10	Ether0		Ether10	Ether0
R1	Internet			
Ether1				

4. Infraestrutura de Roteamento e Transporte:

Para roteamento dinâmico, o OSPF foi implementado no Backbone, tendo como Router ID o endereço de Loopback. Dessa forma, é garantido uma melhor gerência da rede e uma conectividade estável, pois, a interface lógica de Loopback não ficará Down.

- **OSPF Adjacências:**

```
[admin@R1] > routing/ospf/neighbor/print
```

Flags: V - virtual; D - dynamic

```
0 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.65
  router-id=10.100.0.11 state="Full" state-changes=5 adjacency=3m31s
  timeout=39s

1 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.30
  router-id=10.100.0.2 state="Full" state-changes=4 adjacency=2m57s
  timeout=34s

2 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.61
  router-id=10.100.0.12 state="Full" state-changes=5 adjacency=3m31s
  timeout=39s

3 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.34
  router-id=10.100.0.5 state="Full" state-changes=5 adjacency=3m33s
  timeout=37s

4 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.2
  router-id=10.100.0.6 state="Full" state-changes=5 adjacency=3m33s
  timeout=37s

5 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.6
  router-id=10.100.0.9 state="Full" state-changes=5 adjacency=3m36s
  timeout=34s

6 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.53
  router-id=10.100.0.8 state="Full" state-changes=10 adjacency=2m11s
  timeout=35s
```

```
[admin@R2] > routing/ospf/neighbor/print
```

Flags: V - virtual; D - dynamic

- 0 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.17
router-id=10.100.0.6 state="Full" state-changes=5 adjacency=4h47m37s
timeout=34s

- 1 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.13
router-id=10.100.0.3 state="Full" state-changes=6 adjacency=4h17m44s
timeout=33s

- 2 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.22
router-id=10.100.0.10 state="Full" state-changes=6 adjacency=4h17m44s
timeout=31s

- 3 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.26
router-id=10.100.0.13 state="Full" state-changes=6 adjacency=4h17m43s
timeout=38s

- 4 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.29
router-id=10.100.0.1 state="Full" state-changes=4 adjacency=6m51s
timeout=39s

```
[admin@R3] > routing/ospf/neighbor/print
```

Flags: V - virtual; D - dynamic

- 0 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.14
router-id=10.100.0.2 state="Full" state-changes=5 adjacency=4h18m29s
timeout=34s

- 1 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.9
router-id=10.100.0.5 state="Full" state-changes=5 adjacency=4h18m25s
timeout=38s

```
[admin@R4] > routing/ospf/neighbor/print
```

Flags: V - virtual; D - dynamic

- 0 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.46
router-id=10.100.0.7 state="Full" state-changes=6 adjacency=4h19m36s
timeout=34s
- 1 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.42
router-id=10.100.0.6 state="Full" state-changes=5 adjacency=4h19m37s
timeout=34s

```
[admin@R5] > routing/ospf/neighbor/print
```

Flags: V - virtual; D - dynamic

- 0 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.10
router-id=10.100.0.3 state="Full" state-changes=6 adjacency=4h20m6s
timeout=37s
- 1 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.33
router-id=10.100.0.1 state="Full" state-changes=11 adjacency=9m53s
timeout=33s

```
[admin@R6] > routing/ospf/neighbor/print
```

Flags: V - virtual; D - dynamic

- 0 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.18
router-id=10.100.0.2 state="Full" state-changes=6 adjacency=4h50m36s
timeout=31s
- 1 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.1
router-id=10.100.0.1 state="Full" state-changes=11 adjacency=10m24s
timeout=31s
- 2 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.41
router-id=10.100.0.4 state="Full" state-changes=6 adjacency=4h20m45s
timeout=35s

```
[admin@R7] > routing/ospf/neighbor/print
```

```
Flags: V - virtual; D - dynamic
```

```
0 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.50
  router-id=10.100.0.8 state="Full" state-changes=11 adjacency=9m50s
  timeout=31s

1 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.45
  router-id=10.100.0.4 state="Full" state-changes=5 adjacency=4h21m39s
  timeout=40s
```

```
[admin@R8] > routing/ospf/neighbor/print
```

```
Flags: V - virtual; D - dynamic
```

```
0 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.49
  router-id=10.100.0.7 state="Full" state-changes=5 adjacency=10m59s
  timeout=31s

1 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.54
  router-id=10.100.0.1 state="Full" state-changes=5 adjacency=11m4s
  timeout=36s
```

```
[admin@R9] > routing/ospf/neighbor/print
```

```
Flags: V - virtual; D - dynamic
```

```
0 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.5
  router-id=10.100.0.1 state="Full" state-changes=10 adjacency=13m48s
  timeout=31s
```

```
[admin@R10] > routing/ospf/neighbor/print
```

```
Flags: V - virtual; D - dynamic
```

```
0 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.21
  router-id=10.100.0.2 state="Full" state-changes=5 adjacency=4h24m45s
  timeout=39s
```



```
[admin@R11] > routing/ospf/neighbor/print
```

Flags: V - virtual; D - dynamic

```
0 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.66
  router-id=10.100.0.1 state="Full" state-changes=11 adjacency=15m9s
  timeout=35s
```

```
[admin@R12] > routing/ospf/neighbor/print
```

Flags: V - virtual; D - dynamic

```
0 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.62
  router-id=10.100.0.1 state="Full" state-changes=11 adjacency=15m39s
  timeout=35s
```

```
[admin@R13] > routing/ospf/neighbor/print
```

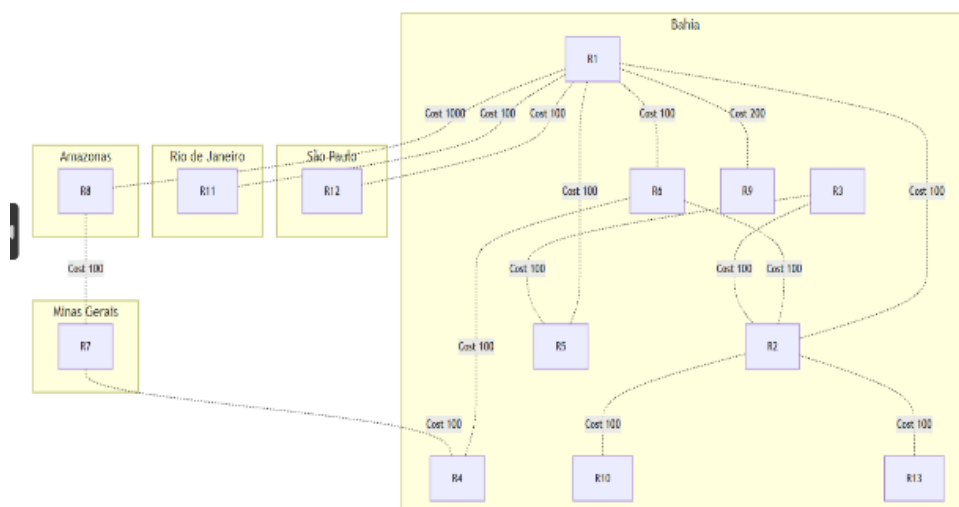
Flags: V - virtual; D - dynamic

```
0 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.25
  router-id=10.100.0.2 state="Full" state-changes=5 adjacency=4h26m23s
  timeout=30s
```

- **Validação dos custos de interface**

O enlace R1-R8, com custo 1000, permanece como o caminho de menor preferência, dando a possibilidade de validar a seleção de caminhos e a redundância.

Os custos exigidos para o projeto estão listados na imagem abaixo:



```
[admin@R1] > routing/ospf/interface-template/print
```

```
Flags: X - disabled, I - inactive
```

```
0  area=ospf-area-1 interfaces=loopback,ether9-Nginx instance-id=0
    type=broadcast retransmit-interval=5s transmit-delay=1s
    hello-interval=10s dead-interval=40s priority=128 cost=1 passive
    auth=md5 auth-key=
```

```
1  ;;; DEMAIS
```

```
    area=ospf-area-1
    interfaces=ether2-R6,ether4-R5,ether5-R2,ether6-R12,ether7-R11
    instance-id=0 type=ptp retransmit-interval=5s transmit-delay=1s
    hello-interval=10s dead-interval=40s priority=128 cost=100 auth=md5
    auth-key=
```

```
2  ;;; R9
```

```
    area=ospf-area-1 interfaces=ether3-R9 instance-id=0 type=ptp
    retransmit-interval=5s transmit-delay=1s hello-interval=10s
    dead-interval=40s priority=128 cost=200 auth=md5 auth-key=
```

```
3  ;;; R8
```

```
    area=ospf-area-1 interfaces=ether8-R8 instance-id=0 type=ptp
    retransmit-interval=5s transmit-delay=1s hello-interval=10s
    dead-interval=40s priority=128 cost=1000 auth=md5 auth-key=
```

```
[admin@R2] > routing/ospf/interface-template/print
```

```
Flags: X - disabled, I - inactive
```

```
0 area=ospf-area-1 interfaces=loopback instance-id=0 type=broadcast
```

```
retransmit-interval=5s transmit-delay=1s hello-interval=10s
```

```
dead-interval=40s priority=128 cost=1 passive auth=md5
```

```
auth-key=
```

```
1 ;;; DEMAIS
```

```
area=ospf-area-1
```

```
interfaces=ether1-R6,ether2-R3,ether3-R10,ether4-R13,ether5-R1
```

```
instance-id=0 type=ptp retransmit-interval=5s transmit-delay=1s
```

```
hello-interval=10s dead-interval=40s priority=128 cost=100 auth=md5
```

```
auth-key=
```

```
[admin@R3] > routing/ospf/interface-template/print
```

```
Flags: X - disabled, I - inactive
```

```
0 area=ospf-area-1 interfaces=loopback instance-id=0 type=broadcast
```

```
retransmit-interval=5s transmit-delay=1s hello-interval=10s
```

```
dead-interval=40s priority=128 cost=1 passive auth=md5
```

```
auth-key=
```

```
1 area=ospf-area-1 interfaces=ether2-R2,ether3-R5 instance-id=0 type=ptp
```

```
retransmit-interval=5s transmit-delay=1s hello-interval=10s
```

```
dead-interval=40s priority=128 cost=100 auth=md5 auth-key=
```

```
[admin@R4] > routing/ospf/interface-template/print
```

```
Flags: X - disabled, I - inactive
```

```
0 area=ospf-area-1 interfaces=loopback instance-id=0 type=broadcast
```

```
retransmit-interval=5s transmit-delay=1s hello-interval=10s
```

```
dead-interval=40s priority=128 cost=1 passive auth=md5
```

```
auth-key=
```

```
1 ;;; DEMAIS
```

```
area=ospf-area-1 interfaces=ether2-R7,ether3-R6 instance-id=0 type=ptp
```

```
retransmit-interval=5s transmit-delay=1s hello-interval=10s
```

```
dead-interval=40s priority=128 cost=100 auth=md5 auth-key=
```

```
[admin@R5] > routing/ospf/interface-template/print
```

```
Flags: X - disabled, I - inactive
```

```
0 area=ospf-area-1 interfaces=loopback instance-id=0 type=broadcast
  retransmit-interval=5s transmit-delay=1s hello-interval=10s
  dead-interval=40s priority=128 cost=1 passive auth=md5
  auth-key=
```

```
1 ;;; DEMAIS
```

```
area=ospf-area-1 interfaces=ether3-R3,ether4-R1 instance-id=0 type=ptp
retransmit-interval=5s transmit-delay=1s hello-interval=10s
dead-interval=40s priority=128 cost=100 auth=md5 auth-key=
```

```
[admin@R6] > routing/ospf/interface-template/print
```

```
Flags: X - disabled, I - inactive
```

```
0 area=ospf-area-1 interfaces=loopback instance-id=0 type=broadcast
  retransmit-interval=5s transmit-delay=1s hello-interval=10s
  dead-interval=40s priority=128 cost=1 passive auth=md5
  auth-key=
```

```
1 ;;; DEMAIS
```

```
area=ospf-area-1 interfaces=ether1-R2,ether2-R1,ether3-R4 instance-id=0
type=ptp retransmit-interval=5s transmit-delay=1s hello-interval=10s
dead-interval=40s priority=128 cost=100 auth=md5 auth-key=
```

```
[admin@R7] > routing/ospf/interface-template/print
```

```
Flags: X - disabled, I - inactive
```

```
0 area=ospf-area-1 interfaces=loopback instance-id=0 type=broadcast
  retransmit-interval=5s transmit-delay=1s hello-interval=10s
  dead-interval=40s priority=128 cost=1 passive auth=md5
  auth-key=
```

```
1 ;;; DEMAIS
```

```
area=ospf-area-1 interfaces=ether1-R8,ether2-R4 instance-id=0 type=ptp
retransmit-interval=5s transmit-delay=1s hello-interval=10s
dead-interval=40s priority=128 cost=100 auth=md5 auth-key=
```

```
[admin@R8] > routing/ospf/interface-template/print
```

```
Flags: X - disabled, I - inactive
```

```
0 area=ospf-area-1 interfaces=loopback instance-id=0 type=broadcast
  retransmit-interval=5s transmit-delay=1s hello-interval=10s
  dead-interval=40s priority=128 cost=1 passive auth=md5
  auth-key=
```

```
1 ;; R7
```

```
area=ospf-area-1 interfaces=ether1-R7 instance-id=0 type=ptp
retransmit-interval=5s transmit-delay=1s hello-interval=10s
dead-interval=40s priority=128 cost=100 auth=md5 auth-key=
```

```
2 ;; R1
```

```
area=ospf-area-1 interfaces=ether8-R1 instance-id=0 type=ptp
retransmit-interval=5s transmit-delay=1s hello-interval=10s
dead-interval=40s priority=128 cost=1000 auth=md5 auth-key=
```

```
[admin@R9] > routing/ospf/interface-template/print
```

```
Flags: X - disabled, I - inactive
```

```
0 area=ospf-area-1 interfaces=loopback instance-id=0 type=broadcast
  retransmit-interval=5s transmit-delay=1s hello-interval=10s
  dead-interval=40s priority=128 cost=1 passive auth=md5
  auth-key=
```

```
1 ;; R1
```

```
area=ospf-area-1 interfaces=ether3-R1 instance-id=0 type=ptp
retransmit-interval=5s transmit-delay=1s hello-interval=10s
dead-interval=40s priority=128 cost=200 auth=md5 auth-key=
```

```
[admin@R10] > routing/ospf/interface-template/print
```

```
Flags: X - disabled, I - inactive
```

```
0 area=ospf-area-1 interfaces=loopback instance-id=0 type=broadcast
```

```
retransmit-interval=5s transmit-delay=1s hello-interval=10s
```

```
dead-interval=40s priority=128 cost=1 passive auth=md5
```

```
auth-key=
```

```
1 ;; R2
```

```
area=ospf-area-1 interfaces=ether3-R2 instance-id=0 type=ptp
```

```
retransmit-interval=5s transmit-delay=1s hello-interval=10s
```

```
dead-interval=40s priority=128 cost=100 auth=md5 auth-key=
```

```
[admin@R11] > routing/ospf/interface-template/print
```

```
Flags: X - disabled, I - inactive
```

```
0 area=ospf-area-1 interfaces=loopback instance-id=0 type=broadcast
```

```
retransmit-interval=5s transmit-delay=1s hello-interval=10s
```

```
dead-interval=40s priority=128 cost=1 passive auth=md5
```

```
auth-key="@Dm1122ti#"
```

```
1 ;; R1
```

```
area=ospf-area-1 interfaces=ether7-R1 instance-id=0 type=ptp
```

```
retransmit-interval=5s transmit-delay=1s hello-interval=10s
```

```
dead-interval=40s priority=128 cost=100 auth=md5 auth-key=
```

```
[admin@R12] > routing/ospf/interface-template/print
```

```
Flags: X - disabled, I - inactive
```

```
0 area=ospf-area-1 interfaces=loopback instance-id=0 type=broadcast
```

```
retransmit-interval=5s transmit-delay=1s hello-interval=10s
```

```
dead-interval=40s priority=128 cost=1 passive auth=md5
```

```
auth-key=
```

```
1 ;; R1
```

```
area=ospf-area-1 interfaces=ether6-R1 instance-id=0 type=ptp
```

```
retransmit-interval=5s transmit-delay=1s hello-interval=10s
```

```
dead-interval=40s priority=128 cost=100 auth=md5 auth-key=
```

```
[admin@R13] > routing/ospf/interface-template/print
```

Flags: X - disabled, I - inactive

```
0 area=ospf-area-1 interfaces=loopback instance-id=0 type=broadcast
```

```
retransmit-interval=5s transmit-delay=1s hello-interval=10s
```

```
dead-interval=40s priority=128 cost=1 passive auth=md5
```

```
auth-key=
```

```
1 ;; R2
```

```
area=ospf-area-1 interfaces=ether4-R2 instance-id=0 type=ptp
```

```
retransmit-interval=5s transmit-delay=1s hello-interval=10s
```

```
dead-interval=40s priority=128 cost=100 auth=md5 auth-key=
```

• MPLS

O MPLS foi habilitado sobre a infraestrutura do OSPF e utiliza como Endereço de Transporte a Loopback, dessa forma, é garantido uma melhor gerência da infraestrutura e evita perdas de conectividade causados por problemas nos enlaces físicos.

Flags MPLS

D	Relação criada dinamicamente
O	Operante
T	Troca de mensagem Unicast com R8
V	Conexão VPLS
P	Atuando como passivo

R1

admin@192.168.18.98 (R1) - WinBox (64bit) v7.2.3 on CHR (x86_64)

Session Settings Dashboard

Safe Mode Session: 192.168.18.98

Quick Set MPLS

CAPsMAN LDP Instance LDP Interface LDP Neighbor Accept Filter Advertise Filter Local Mapping Remote Mapping MPLS Interface

#	Transport	Send ...	Peer	Local Transport	Addresses
0 DOp	10.100.0.8		10.100.0.8:0	10.100.0.1	10.100.0.8, 100.100.1.50, 100.100.1.53, 192.168.10.1
1 DOp	10.100.0.6		10.100.0.6:0	10.100.0.1	10.100.0.6, 100.100.1.2, 100.100.1.17, 100.100.1.42
2 DOp	10.100.0.9		10.100.0.9:0	10.100.0.1	10.100.0.9, 100.100.1.6
3 DOp	10.100.0.5		10.100.0.5:0	10.100.0.1	10.100.0.5, 100.100.1.9, 100.100.1.34
4 DOp	10.100.0.2		10.100.0.2:0	10.100.0.1	10.100.0.2, 100.100.1.14, 100.100.1.18, 100.100.1.21, 100.100.1.25, 100.100.1.30
5 DOp	10.100.0.11		10.100.0.11:0	10.100.0.1	10.100.0.11, 100.100.1.65
6 DOp	10.100.0.12		10.100.0.12:0	10.100.0.1	10.100.0.12, 100.100.1.61

R2

admin@50:00:00:03:00:00 (R2) via 192.168.18.98 - WinBox (64bit) v7.2.3 on CHR (x86_64)

Session Settings Dashboard

Safe Mode Session: 50:00:00:03:00:00

Quick Set MPLS

CAPsMAN LDP Instance LDP Interface LDP Neighbor Accept Filter Advertise Filter Local Mapping Remote Mapping MPLS Interface

#	Transport	Send ...	Peer	Local Transport	Addresses
0 DOp	10.100.0.6		10.100.0.6:0	10.100.0.2	10.100.0.6, 100.100.1.2, 100.100.1.17, 100.100.1.42
1 DOp	10.100.0.3		10.100.0.3:0	10.100.0.2	10.100.0.3, 100.100.1.10, 100.100.1.13
2 DOp	10.100.0.10		10.100.0.10:0	10.100.0.2	10.100.0.10, 100.100.1.22
3 DOp	10.100.0.13		10.100.0.13:0	10.100.0.2	10.100.0.13, 100.100.1.26
4 DOp	10.100.0.1		10.100.0.1:0	10.100.0.2	10.100.0.1, 100.100.1.1, 100.100.1.5, 100.100.1.29, 100.100.1.33, 100.100.1.54, 100.100.1.62, 100.100.1.66, 100.100.1.253, 192.168.10.254, 192.168.18.98

R3

admin@50:00:00:04:00:00 (R3) via 192.168.18.98 - WinBox (64bit) v7.2.3 on CHR (x86_64)

Session Settings Dashboard

Safe Mode Session: 50:00:00:04:00:00

MPLS						
LDP Instance LDP Interface LDP Neighbor Accept Filter Advertise Filter Local Mapping Remote Mapping MPLS Interface						
+ - ✓ ✕ 📄 🔍						
#	Transport	Send ...	Peer	Local Transport	Addresses	
0 DO	10.100.0.2		10.100.0.2:0	10.100.0.3	10.100.0.2, 100.100.1.14, 100.100.1.18, 100.100.1.21, 100.100.1.25, 100.100.1.30	
1 DOp	10.100.0.5		10.100.0.5:0	10.100.0.3	10.100.0.5, 100.100.1.9, 100.100.1.34	

R4

admin@50:00:00:0D:00:00 (R4) via 192.168.18.98 - WinBox (64bit) v7.2.3 on CHR (x86_64)

Session Settings Dashboard

Safe Mode Session: 50:00:00:0D:00:00

MPLS						
LDP Instance LDP Interface LDP Neighbor Accept Filter Advertise Filter Local Mapping Remote Mapping MPLS Interface						
+ - ✓ ✕ 📄 🔍						
#	Transport	Send ...	Peer	Local Transport	Addresses	
0 DOp	10.100.0.7		10.100.0.7:0	10.100.0.4	10.100.0.7, 100.100.1.46, 100.100.1.49	
1 DOp	10.100.0.6		10.100.0.6:0	10.100.0.4	10.100.0.6, 100.100.1.2, 100.100.1.17, 100.100.1.42	

R5

admin@50:00:00:08:00:00 (R5) via 192.168.18.98 - WinBox (64bit) v7.2.3 on CHR (x86_64)

Session Settings Dashboard

Safe Mode Session: 50:00:00:08:00:00

MPLS						
LDP Instance LDP Interface LDP Neighbor Accept Filter Advertise Filter Local Mapping Remote Mapping MPLS Interface						
+ - ✓ ✕ 📄 🔍						
#	Transport	Send ...	Peer	Local Transport	Addresses	
0 DO	10.100.0.3		10.100.0.3:0	10.100.0.5	10.100.0.3, 100.100.1.10, 100.100.1.13	
1 DO	10.100.0.1		10.100.0.1:0	10.100.0.5	10.100.0.1, 100.100.1.1, 100.100.1.5, 100.100.1.29, 100.100.1.33, 100.100.1.54, 100.100.1.62, 100.100.1.66, 100.100.1.253, 192.168.10.254, 192.168.18.98	

R6

admin@50:00:00:05:00:00 (R6) via 192.168.18.98 - WinBox (64bit) v7.2.3 on CHR (x86_64)

Session Settings Dashboard

Safe Mode Session: 50:00:00:05:00:00

MPLS						
LDP Instance LDP Interface LDP Neighbor Accept Filter Advertise Filter Local Mapping Remote Mapping MPLS Interface						
+ - ✓ ✕ 📄 🔍						
#	Transport	Send ...	Peer	Local Transport	Addresses	
0 DO	10.100.0.2		10.100.0.2:0	10.100.0.6	10.100.0.2, 100.100.1.14, 100.100.1.18, 100.100.1.21, 100.100.1.25, 100.100.1.30	
1 DO	10.100.0.4		10.100.0.4:0	10.100.0.6	10.100.0.4, 100.100.1.41, 100.100.1.45	
2 DO	10.100.0.1		10.100.0.1:0	10.100.0.6	10.100.0.1, 100.100.1.1, 100.100.1.5, 100.100.1.29, 100.100.1.33, 100.100.1.54, 100.100.1.62, 100.100.1.66, 100.100.1.253, 192.168.10.254, 192.168.18.98	

R7

admin@50:00:00:0B:00:00 (R7) via 192.168.18.98 - WinBox (64bit) v7.2.3 on CHR (x86_64)

Session Settings Dashboard

Safe Mode Session: 50:00:00:0B:00:00

MPLS						
LDP Instance LDP Interface LDP Neighbor Accept Filter Advertise Filter Local Mapping Remote Mapping MPLS Interface						
+ - ✓ ✕ 📄 🔍						
#	Transport	Send ...	Peer	Local Transport	Addresses	
0 DO	10.100.0.4		10.100.0.4:0	10.100.0.7	10.100.0.4, 100.100.1.41, 100.100.1.45	
1 DOp	10.100.0.8		10.100.0.8:0	10.100.0.7	10.100.0.8, 100.100.1.50, 100.100.1.53, 192.168.10.1	

R8

admin@50:00:00:06:00:00 (R8) via 192.168.18.98 - WinBox (64bit) v7.2.3 on CHR (x86_64)

Session Settings Dashboard

Safe Mode Session: 50:00:00:06:00:00

MPLS						
LDP Instance LDP Interface LDP Neighbor Accept Filter Advertise Filter Local Mapping Remote Mapping MPLS Interface						
+ - ✓ ✕ 📄 🔍						
#	Transport	Send ...	Peer	Local Transport	Addresses	
0 DOp	10.100.0.1		10.100.0.1:0	10.100.0.8	10.100.0.1, 100.100.1.1, 100.100.1.5, 100.100.1.29, 100.100.1.33, 100.100.1.54, 100.100.1.62, 100.100.1.66, 100.100.1.253, 192.168.10.254, 192.168.18.98	
1 DO	10.100.0.7		10.100.0.7:0	10.100.0.8	10.100.0.7, 100.100.1.46, 100.100.1.49	

R9

admin@50:00:00:02:00:00 (R9) via 192.168.18.98 - WinBox (64bit) v7.2.3 on CHR (x86_64)

Session Settings Dashboard

Safe Mode Session: 50:00:00:02:00:00

Quick Set MPLS

LDP Instance LDP Interface LDP Neighbor Accept Filter Advertise Filter Local Mapping Remote Mapping MPLS Interface

#	Transport	Send ...	Peer	Local Transport	Addresses
0 DO	10.100.0.1		10.100.0.1.0	10.100.0.9	10.100.0.1, 100.100.1.1, 100.100.1.5, 100.100.1.29, 100.100.1.33, 100.100.1.54, 100.100.1.62, 100.100.1.66, 100.100.1.253, 192.168.10.254, 192.168.18.98

R10

admin@50:00:00:0A:00:00 (R10) via 192.168.18.98 - WinBox (64bit) v7.2.3 on CHR (x86_64)

Session Settings Dashboard

Safe Mode Session: 50:00:00:0A:00:00

Quick Set MPLS

LDP Instance LDP Interface LDP Neighbor Accept Filter Advertise Filter Local Mapping Remote Mapping MPLS Interface

#	Transport	Send ...	Peer	Local Transport	Addresses
0 DO	10.100.0.2		10.100.0.2.0	10.100.0.10	10.100.0.2, 100.100.1.14, 100.100.1.18, 100.100.1.21, 100.100.1.25, 100.100.1.30

R11

admin@50:00:00:07:00:00 (R11) via 192.168.18.98 - WinBox (64bit) v7.2.3 on CHR (x86_64)

Session Settings Dashboard

Safe Mode Session: 50:00:00:07:00:00

Quick Set MPLS

LDP Instance LDP Interface LDP Neighbor Accept Filter Advertise Filter Local Mapping Remote Mapping MPLS Interface

#	Transport	Send ...	Peer	Local Transport	Addresses
0 DO	10.100.0.1		10.100.0.1.0	10.100.0.11	10.100.0.1, 100.100.1.1, 100.100.1.5, 100.100.1.29, 100.100.1.33, 100.100.1.54, 100.100.1.62, 100.100.1.66, 100.100.1.253, 192.168.10.254, 192.168.18.98

R12

admin@50:00:00:0C:00:00 (R12) via 192.168.18.98 - WinBox (64bit) v7.2.3 on CHR (x86_64)

Session Settings Dashboard

Safe Mode Session: 50:00:00:0C:00:00

Quick Set MPLS

LDP Instance LDP Interface LDP Neighbor Accept Filter Advertise Filter Local Mapping Remote Mapping MPLS Interface

#	Transport	Send ...	Peer	Local Transport	Addresses
0 DO	10.100.0.1		10.100.0.1.0	10.100.0.12	10.100.0.1, 100.100.1.1, 100.100.1.5, 100.100.1.29, 100.100.1.33, 100.100.1.54, 100.100.1.62, 100.100.1.66, 100.100.1.253, 192.168.10.254, 192.168.18.98

R13

admin@50:00:00:09:00:00 (R13) via 192.168.18.98 - WinBox (64bit) v7.2.3 on CHR (x86_64)

Session Settings Dashboard

Safe Mode Session: 50:00:00:09:00:00

Quick Set MPLS

LDP Instance LDP Interface LDP Neighbor Accept Filter Advertise Filter Local Mapping Remote Mapping MPLS Interface

#	Transport	Send ...	Peer	Local Transport	Addresses
0 DO	10.100.0.2		10.100.0.2.0	10.100.0.13	10.100.0.2, 100.100.1.14, 100.100.1.18, 100.100.1.21, 100.100.1.25, 100.100.1.30

• Túnel VPLS

Para a conexão L2 entre R1 e R8, foi criada uma interface VPLS em cada lado. Como *Remote Peer*, foram utilizados os endereços de Loopback dos roteadores, garantindo a comunicação L2 independentemente de quaisquer eventualidades nos enlaces físicos. Nos dois lados, a interface VPLS foi adicionada a uma Bridge juntamente com a interface física ligada ao VPC (ether10-VPC). No lado do R8, a Bridge fornece endereçamento IP aos dispositivos pertencentes ao domínio L2 via DHCP Server. No lado do R1, a Bridge recebe essas informações via DHCP Client, permitindo que a VPC-R1 consiga endereçamento.

➤ Interfaces VPLS R1 e R8

```
[admin@R1] > interface/vpls/print
```

```
Flags: R - RUNNING
```

```
Columns: NAME, PEER, VPLS-ID
```

```
# NAME    PEER    VPLS-ID
```

```
0 R vpls-R8 10.100.0.8 1:1
```

```
[admin@R8] > interface/vpls/print
```

```
Flags: R - RUNNING
```

```
Columns: NAME, PEER, VPLS-ID
```

```
# NAME    PEER    VPLS-ID
```

```
0 R vpls-R1 10.100.0.1 1:1
```

➤ Bridge R1 e R8

```
[admin@R1] > interface/bridge/port/print
```

```
Flags: D - DYNAMIC
```

```
Columns: INTERFACE, BRIDGE, HW, PVID, PRIORITY, PATH-COST, INTERNAL-PATH-COST, H  
ORIZON
```

```
# INTERFACE BRIDGE HW PVID PRIORITY PATH-COST IN HORIZON
```

```
0 ether10-VPC VPLS-R8 yes 1 0x80 10 10 none
```

```
1 D vpls-R8 VPLS-R8 1 0x80 10 10 none
```

```
[admin@R8] > interface/bridge/port/print
```

```
Flags: D - DYNAMIC
```

```
Columns: INTERFACE, BRIDGE, HW, PVID, PRIORITY, PATH-COST, INTERNAL-PATH-COST,  
HORIZON
```

```
# INTERFACE BRIDGE HW PVID PRIORITY PATH-COST IN HORIZON
```

```
0 ether10-VPC VPLS-R1 yes 1 0x80 10 10 none
```

```
1 D vpls-R1 VPLS-R1 1 0x80 10 10 none
```

➤ DHCP R1 e R8

```
[admin@R1] > ip dhcp-client/print
```

```
Columns: INTERFACE, USE-PEER-DNS, ADD-DEFAULT-ROUTE, STATUS, ADDRESS
```

```
# INTERFACE USE-PEER-DNS ADD-DEFAULT-ROUTE STATUS ADDRESS
```

```
0 ether1 yes yes bound 192.168.18.98/24
```

```
1 VPLS-R8 no no bound 192.168.10.254/24
```

```
[admin@R8] > ip dhcp-server/print
```

Columns: NAME, INTERFACE, ADDRESS-POOL, LEASE-TIME

```
# NAME  INTERFACE  ADDRESS-POOL  LEASE-TIME
```

```
0 dhcp1  VPLS-R1   dhcp_pool0   10m
```

➤ Tabela de Macs R1 e R8

```
[admin@R1] > interface/bridge/host/print
```

Flags: D - DYNAMIC; L - LOCAL

Columns: MAC-ADDRESS, ON-INTERFACE, BRIDGE, AGE

```
#  MAC-ADDRESS  ON-INTERFACE  BRIDGE  AGE
```

```
0 DL 0E:9C:FB:1B:24:F6 loopback  loopback
```

```
1 D 00:50:79:66:68:0E vpls-R8  VPLS-R8  3m54s
```

```
2 D 00:50:79:66:68:0F ether10-VPC VPLS-R8  3m46s
```

```
3 DL 02:77:DC:C2:17:BB vpls-R8  VPLS-R8
```

```
4 D 02:7F:6B:7B:28:05 vpls-R8  VPLS-R8  11s
```

```
5 DL 50:00:00:01:00:09 ether10-VPC VPLS-R8
```

```
6 D 50:00:00:06:00:09 vpls-R8  VPLS-R8  1m12s
```

```
[admin@R8] > interface/bridge/host/print
```

Flags: D - DYNAMIC; L - LOCAL

Columns: MAC-ADDRESS, ON-INTERFACE, BRIDGE, AGE

```
#  MAC-ADDRESS  ON-INTERFACE  BRIDGE  AGE
```

```
0 DL 86:7D:EE:34:DD:4F loopback  loopback
```

```
1 D 00:50:79:66:68:0E ether10-VPC VPLS-R1  4m3s
```

```
2 D 00:50:79:66:68:0F vpls-R1  VPLS-R1  3m55s
```

```
3 D 02:77:DC:C2:17:BB vpls-R1  VPLS-R1  2s
```

```
4 DL 02:7F:6B:7B:28:05 vpls-R1  VPLS-R1
```

```
5 D 50:00:00:01:00:09 vpls-R1  VPLS-R1  1m21s
```

```
6 DL 50:00:00:06:00:09 ether10-VPC VPLS-R1
```

➤ Testes de comunicação entre VPC-R1 e VPC-R8

```
VPC-R1
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
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Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS> dhcp
DORA IP 192.168.10.253/24 GW 192.168.10.1

VPCS> ping 192.168.10.252

64 bytes from 192.168.10.252 icmp_seq=1 ttl=64 time=2.616 ms
64 bytes from 192.168.10.252 icmp_seq=2 ttl=64 time=2.885 ms
64 bytes from 192.168.10.252 icmp_seq=3 ttl=64 time=3.033 ms
64 bytes from 192.168.10.252 icmp_seq=4 ttl=64 time=3.313 ms
64 bytes from 192.168.10.252 icmp_seq=5 ttl=64 time=2.877 ms

VPCS> 
```

```
VPC-R8
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
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Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS> dhcp
DORA IP 192.168.10.252/24 GW 192.168.10.1

VPCS> ping 192.168.10.253

64 bytes from 192.168.10.253 icmp_seq=1 ttl=64 time=2.348 ms
64 bytes from 192.168.10.253 icmp_seq=2 ttl=64 time=2.607 ms
64 bytes from 192.168.10.253 icmp_seq=3 ttl=64 time=3.105 ms
64 bytes from 192.168.10.253 icmp_seq=4 ttl=64 time=2.568 ms
64 bytes from 192.168.10.253 icmp_seq=5 ttl=64 time=2.851 ms

VPCS> 
```

5. Servidor de aplicação, Nginx

Simulando uma aplicação, seguindo os requisitos, um servidor Linux com Nginx instalado foi adicionado na topologia, estando acessível por toda a rede. O servidor está conectado diretamente no R1, pela interface física ether9, sendo adicionada no OSPF para a distribuição do acesso aos demais equipamentos do Backbone.

➤ Compartilhamento de rota padrão via OSPF

O OSPF já compartilha uma rota padrão, o que já possibilitaria a conectividade dos equipamentos com o servidor, mas, por boa prática, a rede do servidor foi adicionada no roteamento dinâmico em modo passivo, juntamente com a Loopback.

```
[admin@R1] > routing/ospf/instance/print  
  
Flags: X - disabled, I - inactive  
  
0  name="ospf-instance-1" version=2 vrf=main router-id=10.100.0.1  
    originate-default=always redistribute=""
```

```
[admin@R1] > routing/ospf/interface-template/print  
  
Flags: X - disabled, I - inactive  
  
0  area=ospf-area-1 interfaces=loopback, ether9-Nginx instance-id=0  
    type=broadcast retransmit-interval=5s transmit-delay=1s  
    hello-interval=10s dead-interval=40s priority=128 cost=1 passive  
    auth=md5 auth-key="@Dm1122ti#"
```

➤ Teste de comunicação do R7 com o Nginx

```
[admin@R7] > /tool fetch url="http://100.100.1.254" dst-path=index.html  
  
status: finished  
  
downloaded: 0KiB-c-z pause]  
  
total: 0KiB  
  
duration: 1s  
  
[admin@R7] > file/print where name~"index.html"  
  
Columns: NAME, TYPE, SIZE, CREATION-TIME  
  
# NAME      TYPE      SIZE  CREATION-TIME  
0 index.html .html file  612  sep/02/2026 16:23:58
```

➤ Página HTML Nginx

file:///C:/Users/Juin/Downloads/index.html

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

6.Testes específicos solicitados R3 <-> R7

➤ PING

```
R3
[admin@R3] > ip address/print
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS      NETWORK      INTERFACE
;; LOOPBACK
0 10.100.0.3/32  10.100.0.3   loopback
;; PARA-R5
1 100.100.1.10/30 100.100.1.8  ether3-R5
;; PARA-R2
2 100.100.1.13/30 100.100.1.12 ether2-R2
[admin@R3] > ping 10.100.0.7
SEQ HOST      SIZE TTL TIME      STATUS
0 10.100.0.7  56  61 2ms255us
1 10.100.0.7  56  61 2ms241us
2 10.100.0.7  56  61 2ms576us
3 10.100.0.7  56  61 2ms516us
4 10.100.0.7  56  61 2ms241us
5 10.100.0.7  56  61 2ms427us
6 10.100.0.7  56  61 2ms903us
7 10.100.0.7  56  61 2ms345us
8 10.100.0.7  56  61 2ms413us
9 10.100.0.7  56  61 2ms701us
10 10.100.0.7  56  61 3ms571us
11 10.100.0.7  56  61 2ms139us
sent=12 received=12 packet-loss=0% min-rtt=2ms139us avg-rtt=2ms527us
```

```
R7
[admin@R7] > ip address/print
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS      NETWORK      INTERFACE
;; LOOPBACK
0 10.100.0.7/32  10.100.0.7   loopback
;; PARA-R4
1 100.100.1.46/30 100.100.1.44 ether2-R4
;; PARA-R8
2 100.100.1.49/30 100.100.1.48 ether1-R8
[admin@R7] > ping 10.100.0.3
SEQ HOST      SIZE TTL TIME      STATUS
0 10.100.0.3  56  61 2ms534us
1 10.100.0.3  56  61 2ms161us
2 10.100.0.3  56  61 2ms427us
3 10.100.0.3  56  61 2ms654us
4 10.100.0.3  56  61 2ms495us
5 10.100.0.3  56  61 2ms184us
6 10.100.0.3  56  61 2ms739us
7 10.100.0.3  56  61 2ms247us
8 10.100.0.3  56  61 3ms371us
9 10.100.0.3  56  61 2ms447us
10 10.100.0.3  56  61 2ms535us
sent=11 received=11 packet-loss=0% min-rtt=2ms161us avg-rtt=2ms526us
max-rtt=3ms371us
```

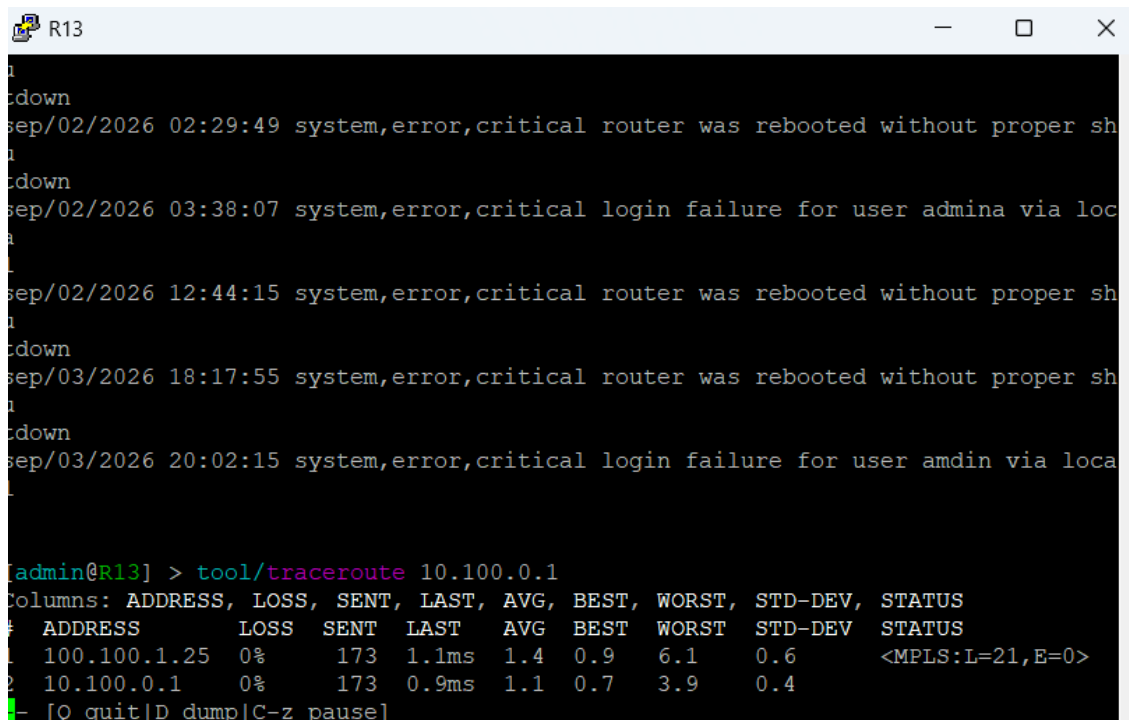
➤ Traceroute

```
[admin@R3] > tool/traceroute 10.100.0.7
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV, STATUS
# ADDRESS      LOSS  SENT  LAST  AVG  BEST  WORST  STD-DEV  STATUS
1 100.100.1.14  0%    23    2.7ms 2.7  1.7   7.4    1.1     <MPLS:L=22,E=0>
2 100.100.1.17  0%    23    2.1ms 2.6  1.7   6.9    1.1     <MPLS:L=24,E=0>
3 100.100.1.41  0%    23    2.1ms 2.2  1.6   3.4    0.5     <MPLS:L=21,E=0>
4 10.100.0.7    0%    23     2ms  2.3  1.6   7      1.2
- [Q quit|D dump|C-z pause]
```

```
[admin@R7] > tool/traceroute 10.100.0.3
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV, STATUS
# ADDRESS      LOSS  SENT  LAST  AVG  BEST  WORST  STD-DEV  STATUS
1 100.100.1.45  0%     2    2.6ms 3.1  2.6   3.6    0.5     <MPLS:L=18,E=0>
2 100.100.1.42  0%     2     2ms  2.4  2     2.7    0.4     <MPLS:L=28,E=0>
3 100.100.1.18  0%     2    2.1ms 2.3  2.1   2.5    0.2     <MPLS:L=24,E=0>
4 10.100.0.3    0%     2    2.1ms 2.3  2.1   2.4    0.2
- [Q quit|D dump|C-z pause]
```

7. Testes de redundância

O teste envolve uma comunicação ICMP do Router R13 para o R1. Como resultado, nas imagens abaixo, será mostrado o caminho inicial usado até o endereço do R1, o impacto após a queda do caminho principal e a restauração da comunicação por meio de outro caminho.



```
R13
down
sep/02/2026 02:29:49 system,error,critical router was rebooted without proper sh
down
sep/02/2026 03:38:07 system,error,critical login failure for user admina via loc
sep/02/2026 12:44:15 system,error,critical router was rebooted without proper sh
down
sep/03/2026 18:17:55 system,error,critical router was rebooted without proper sh
down
sep/03/2026 20:02:15 system,error,critical login failure for user amdin via loca

[admin@R13] > tool/traceroute 10.100.0.1
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV, STATUS
# ADDRESS      LOSS  SENT  LAST  AVG  BEST  WORST  STD-DEV  STATUS
1 100.100.1.25  0%   173   1.1ms 1.4  0.9   6.1    0.6     <MPLS:L=21,E=0>
2 10.100.0.1    0%   173   0.9ms 1.1  0.7   3.9    0.4
- [Q quit|D dump|C-z pause]
```

- Interface ether4 Down (Caminho Principal)

```

R1
Columns: NAME, TYPE, ACTUAL-MTU, L2MTU, MAC-ADDRESS
# NAME TYPE ACTUAL-MTU L2MTU MAC-ADDRESS
;;; NET
0 R ether1 ether 1500 50:00:00:01:00:00
;;; R6
1 R ether2-R6 ether 9000 50:00:00:01:00:01
;;; R9
2 R ether3-R9 ether 1500 50:00:00:01:00:02
;;; R5
3 R ether4-R5 ether 1500 50:00:00:01:00:03
;;; R2
4 R ether5-R2 ether 9000 50:00:00:01:00:04
;;; R12
5 R ether6-R12 ether 1500 50:00:00:01:00:05
;;; R11
6 R ether7-R11 ether 1500 50:00:00:01:00:06
;;; R8
7 R ether8-R8 ether 9000 50:00:00:01:00:07
;;; Nginx
8 R ether9-Nginx ether 1500 50:00:00:01:00:08
;;; LAN
9 RS ether10-VPC ether 1500 50:00:00:01:00:09
10 R VPLS-R8 bridge 1500 65535 50:00:00:01:00:09
[admin@R1] > interface/disable number=4

```

- Perda de comunicação com R1

```

R13
u
tdown
sep/02/2026 02:29:49 system,error,critical router was rebooted without proper sh
u
tdown
sep/02/2026 03:38:07 system,error,critical login failure for user admina via loc
a
l
sep/02/2026 12:44:15 system,error,critical router was rebooted without proper sh
u
tdown
sep/03/2026 18:17:55 system,error,critical router was rebooted without proper sh
u
tdown
sep/03/2026 20:02:15 system,error,critical login failure for user amdin via loca
l
[admin@R13] > tool/traceroute 10.100.0.1
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV, STATUS
# ADDRESS LOSS SENT LAST AVG BEST WORST STD-DEV STATUS >
1 100.100.1.25 0.2% 965 timeout 1.4 0.8 23.6 0.9 <MPLS:L=21,E=> >
2 10.100.0.1 0.1% 965 timeout 1.1 0.6 18.3 0.7 >
- [Q quit|D dump|C-z pause|right]

```


- Retomada da conexão por outro caminho

```

R13
sep/02/2026 12:44:15 system,error,critical router was rebooted without proper sh
1
tdown
sep/03/2026 18:17:55 system,error,critical router was rebooted without proper sh
1
tdown
sep/03/2026 20:02:15 system,error,critical login failure for user amdin via loca
1

[admin@R13] > tool/traceroute 10.100.0.1
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV, STATUS
# ADDRESS      LOSS  SENT  LAST  AVG  BEST  WORST  STD-DEV  STATUS
1 100.100.1.25  0.6%  996   1.8ms 1.4   0.8   23.6   0.9      <MPLS:L=21,E=0>
2 10.100.0.1    0.5%  996   1.6ms 1.1   0.6   18.3   0.7      <MPLS:L=35,E=0>
  100.100.1.17

[admin@R13] > tool/traceroute 10.100.0.1
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV, STATUS
# ADDRESS      LOSS  SENT  LAST  AVG  BEST  WORST  STD-DEV  STATUS
1 100.100.1.25  0%    6     1.8ms 2     1.7   2.5    0.3      <MPLS:L=21,E=0>
2 100.100.1.17  0%    6     1.6ms 1.6   1.3   1.8    0.2      <MPLS:L=35,E=0>
3 10.100.0.1    0%    6     1.5ms 1.5   1.3   1.9    0.2
- [Q quit][D dump][C-z pause]

```

- Endereços relacionados

```

R2
tdown
sep/02/2026 12:44:22 system,error,critical router was rebooted without proper sh
u
tdown
sep/03/2026 18:18:04 system,error,critical router was rebooted without proper sh
u
tdown

[admin@R2] > ip address/print
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS      NETWORK      INTERFACE
;;; LOOPBACK
0 10.100.0.2/32  10.100.0.2   loopback
;;; PARA-R3
1 100.100.1.14/30 100.100.1.12 ether2-R3
;;; PARA-R6
2 100.100.1.18/30 100.100.1.16 ether1-R6 ←
;;; PARA-R10
3 100.100.1.21/30 100.100.1.20 ether3-R10
;;; PARA-R13
4 100.100.1.25/30 100.100.1.24 ether4-R13 ←
;;; PARA-R1
5 100.100.1.30/30 100.100.1.28 ether5-R1
[admin@R2] >

```

```
[admin@R6] > ip address/print
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS NETWORK INTERFACE
;;; LOOPBACK
0 10.100.0.6/32 10.100.0.6 loopback
;;; PARA-R1
1 100.100.1.2/30 100.100.1.0 ether2-R1
;;; PARA-R2
2 100.100.1.17/30 100.100.1.16 ether1-R2
;;; PARA-R4
3 100.100.1.42/30 100.100.1.40 ether3-R4
[admin@R6] >
```

- Validação da preferência de rota R8<->R1

```
[admin@R1] > tool/traceroute 10.100.0.8
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV, STATUS
# ADDRESS LOSS SENT LAST AVG BEST WORST STD-DEV STATUS
1 100.100.1.2 0% 48 2ms 3 2 12.7 1.8 <MPLS:L=27,E=0>
2 100.100.1.41 0% 48 1.6ms 2.5 1.6 13.2 1.8 <MPLS:L=22,E=0>
3 100.100.1.46 0% 48 1.6ms 2.4 1.6 20.2 2.7 <MPLS:L=17,E=0>
4 10.100.0.8 0% 48 1.8ms 2.3 1.5 10.7 1.4
[Q quit|D dump|C-z pause]
```

```
[admin@R8] > tool/traceroute 10.100.0.1
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV, STATUS
# ADDRESS LOSS SENT LAST AVG BEST WORST STD-DEV STATUS
1 100.100.1.49 0% 46 3ms 2.6 1.8 6.3 0.7 <MPLS:L=20,E=0>
2 100.100.1.45 0% 46 1.9ms 2.4 1.4 10.6 1.4 <MPLS:L=16,E=0>
3 100.100.1.42 0% 46 1.9ms 2.4 1.5 16.4 2.1 <MPLS:L=35,E=0>
4 10.100.0.1 0% 46 2.1ms 2.3 1.5 14 1.8
[Q quit|D dump|C-z pause]
```

- Queda de interface com menor custo

```
[admin@R7] > interface/p
Flags: R - RUNNING
Columns: NAME, TYPE, ACTUAL-MTU, L2MTU, MAC-ADDRESS
# NAME TYPE ACTUAL-MTU L2MTU MAC-ADDRESS
;;; R8
0 R ether1-R8 ether 9000 50:00:00:0B:00:00
;;; R4
1 R ether2-R4 ether 9000 50:00:00:0B:00:01
2 R ether3 ether 1500 50:00:00:0B:00:02
3 R ether4 ether 1500 50:00:00:0B:00:03
4 R ether5 ether 1500 50:00:00:0B:00:04
5 R ether6 ether 1500 50:00:00:0B:00:05
6 R ether7 ether 1500 50:00:00:0B:00:06
7 R ether8 ether 1500 50:00:00:0B:00:07
8 R ether9 ether 1500 50:00:00:0B:00:08
9 R ether10 ether 1500 50:00:00:0B:00:09
10 R loopback bridge 1500 65535 66:02:67:07:AC:A2
[admin@R7] > interface/disable number=0
```

- Perda de comunicação

```
[admin@R8] > tool/traceroute 10.100.0.1
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV, STATUS
# ADDRESS      LOSS  SENT  LAST    AVG  BEST  WORST  STD-DEV  STATUS
1 100.100.1.49  0.8%  387   timeout 2.8  1.8   13.4   1.3      <MPLS:L=20,E=>
2 100.100.1.45  0.8%  387   timeout 2.3  1.4   10.6   0.9      <MPLS:L=16,E=>
3 100.100.1.42  0.5%  387   timeout 2.2  1.3   16.4   1        <MPLS:L=35,E=>
4 10.100.0.1    0.5%  386   timeout 2.2  1.3   14      1.2      >
[Q quit|D dump|C-z pause|right]
```

```
[admin@R1] > tool/traceroute 10.100.0.8
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV, STATUS
# ADDRESS      LOSS  SENT  LAST    AVG  BEST  WORST  STD-DEV  STATUS
1 100.100.1.2    1.1%  376   timeout 2.7  1.8   12.7   1.1      <MPLS:L=27,E=>
2 100.100.1.41  0.8%  376   timeout 2.3  1.5   13.2   1.1      <MPLS:L=22,E=>
3 100.100.1.46  0.8%  375   timeout 2.2  1.4   20.2   1.2      <MPLS:L=17,E=>
4 10.100.0.8     0.8%  375   timeout 2.1  1.4   12.1   0.9      >
[Q quit|D dump|C-z pause|right]
```

- Restauração pelo caminho de maior custo (1000)

R8 possui uma ligação direta com o R1, porém, no OSPF foi definido o custo de 1000 para o enlace.

```
[admin@R8] > tool/traceroute 10.100.0.1
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV
# ADDRESS      LOSS  SENT  LAST    AVG  BEST  WORST  STD-DEV
1 10.100.0.1    0%      8  0.7ms  0.7  0.5   1.1    0.2
[Q quit|D dump|C-z pause]
```

```
[admin@R1] > tool/traceroute 10.100.0.8
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV
# ADDRESS      LOSS  SENT  LAST    AVG  BEST  WORST  STD-DEV
1 10.100.0.8    0%      18 0.6ms  0.8  0.5   1.6    0.3
[Q quit|D dump|C-z pause]
```

- Endereços relacionados

```
[admin@R7] > ip address/print
Flags: I, D - DYNAMIC
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS      NETWORK      INTERFACE
;;; LOOPBACK
0 10.100.0.7/32  10.100.0.7   loopback
;;; PARA-R4
1 100.100.1.46/30 100.100.1.44 ether2-R4
;;; PARA-R8
2 I 100.100.1.49/30 100.100.1.48 ether1-R8
[admin@R7] >
```

A interface em azul foi desabilitada para o teste

```
[admin@R6] > ip address/print
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS          NETWORK          INTERFACE
;;; LOOPBACK
0 10.100.0.6/32     10.100.0.6      loopback
;;; PARA-R1
1 100.100.1.2/30    100.100.1.0     ether2-R1
;;; PARA-R2
2 100.100.1.17/30   100.100.1.16    ether1-R2
;;; PARA-R4
3 100.100.1.42/30   100.100.1.40    ether3-R4
[admin@R6] >
```

- Caminho principal restabelecido

```
[admin@R7] > interface/enable number=0
[admin@R7] > ip address/print
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS          NETWORK          INTERFACE
;;; LOOPBACK
0 10.100.0.7/32     10.100.0.7      loopback
;;; PARA-R4
1 100.100.1.46/30    100.100.1.44     ether2-R4
;;; PARA-R8
2 100.100.1.49/30    100.100.1.48     ether1-R8
[admin@R7] >
```

Interface habilitada

```
[admin@R1] > tool/traceroute 10.100.0.8
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV, STATUS
# ADDRESS          LOSS  SENT  LAST  AVG  BEST  WORST  STD-DEV  STATUS
1 100.100.1.2        0%    86    2.6ms 3.6  2.1  23.3    3      <MPLS:L=27,E=0>
2 100.100.1.41       0%    86    2ms   2.8  1.6  11.8    1.6     <MPLS:L=22,E=0>
3 100.100.1.46       0%    86    2.2ms 2.7  1.6  10.8    1.5     <MPLS:L=17,E=0>
4 10.100.0.8         0%    86    3.9ms 3    1.6  14.5    2.1
- [Q quit|D dump|C-z pause]
```

Caminho Principal
Restaurado

```
[admin@R8] > tool/traceroute 10.100.0.1
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV, STATUS
# ADDRESS          LOSS  SENT  LAST  AVG  BEST  WORST  STD-DEV  STATUS
1 100.100.1.49       0%    139   6.9ms 2.9  1.4  9.1     1.3     <MPLS:L=20,E=0>
2 100.100.1.45       0%    139   2ms   2.6  1.4  19.9    1.9     <MPLS:L=16,E=0>
3 100.100.1.42       0%    139   2.1ms 2.4  1.4  8.7     1.2     <MPLS:L=35,E=0>
4 10.100.0.1         0%    139   1.9ms 2.4  1.4  13.2    1.3
- [Q quit|D dump|C-z pause]
```

Caminho Principal
Restaurado

8. Troubleshooting

- **State Exchange**

Na verificação final das configurações dos Routers, foi identificado que o state na vizinhança do R1 com R2 estava travado em “Exchange”, após verificações, foi visto uma divergência nos MTUs das interfaces do enlace. Feito a correção, os Routers ficaram com state “Full” novamente.

```
[admin@R1] > routing/ospf/neighbor/print where state~"Exchange"
Flags: V - virtual; D - dynamic
 3 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.30
   router-id=10.100.0.2 state="Exchange" state-changes=4 ls-retransmits=14
   timeout=33s
```

```
[admin@R1] > interface/print where name="ether5"
Flags: R - RUNNING
Columns: NAME, TYPE, ACTUAL-MTU, MAC-ADDRESS
# NAME TYPE ACTUAL-MTU MAC-ADDRESS
;;; R2
0 R ether5 ether 9000 50:00:00:01:00:04
[admin@R1] > []
```

```
[admin@R2] > interface/print where name="ether5"
Flags: R - RUNNING
Columns: NAME, TYPE, ACTUAL-MTU, MAC-ADDRESS
# NAME TYPE ACTUAL-MTU MAC-ADDRESS
;;; R1
0 R ether5 ether 1500 50:00:00:03:00:04
[admin@R2] > []
```

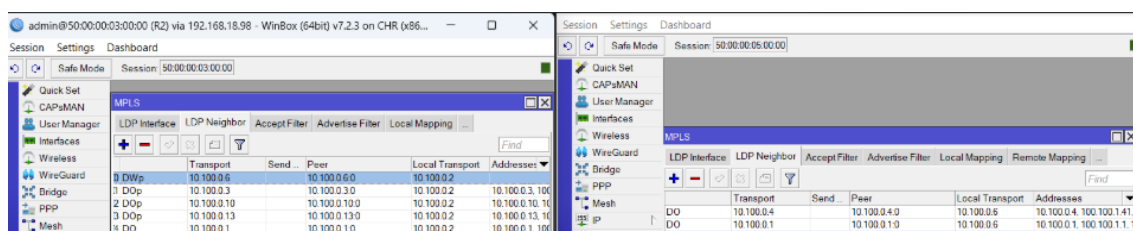
```
[admin@R1] > routing/ospf/neighbor/print where address ~"100.100.1.30"
Flags: V - virtual; D - dynamic
 3 D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.30
   router-id=10.100.0.2 state="Full" state-changes=5 adjacency=1m15s
   timeout=40s
[admin@R1] > []
```

- **Falha no MPLS R2 com R6**

Ainda na validação de configuração dos Routers, foi feito um teste do R2 para o R8 e foi percebido que o MPLS não atuava ao passar pelo R6. A imagem mostra isso claramente com o status nulo, sem a Label MPLS

```
admin@R2] > tool/traceroute 10.100.0.8
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV, STATUS
ADDRESS LOSS SENT LAST AVG BEST WORST STD-DEV STATUS
100.100.1.17 0% 342 0.9ms 1 0.4 28.9 2.3
100.100.1.41 0% 342 2.5ms 2.9 1.4 89.4 5 <MPLS:L=21,E=0>
100.100.1.46 0% 342 2.7ms 2.6 1.4 49.4 3.2 <MPLS:L=19,E=0>
10.100.0.8 0% 342 2.4ms 2.3 1.4 12 1.1
```

Ao visitar as configurações do MPLS no R2 e no R6, foi visto que o campo ADDRESSES, em ambos os Routers, estava sem informação.



Após algumas verificações, chegou o momento de checar os endereços de ambos os Routers. Foi identificado que o endereço da rede de Loopback do R6 estava com o endereço da interface de Loopback do R2, impedindo a comunicação.

```
[admin@R2] > /ip/address/print
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS NETWORK INTERFACE
;;; LOOPBACK
0 10.100.0.2/32 10.100.0.2 loopback
;;; PARA-R3
1 100.100.1.14/30 100.100.1.12 ether2
;;; PARA-R6
2 100.100.1.18/30 100.100.1.16 ether1
;;; PARA-R10
3 100.100.1.21/30 100.100.1.20 ether3
;;; PARA-R13
4 100.100.1.25/30 100.100.1.24 ether4
;;; PARA-R1
5 100.100.1.30/30 100.100.1.28 ether5
[admin@R2] >

[admin@R6] > /ip/address/print
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS NETWORK INTERFACE
;;; LOOPBACK
0 10.100.0.6/32 10.100.0.2 loopback
;;; PARA-R1
1 100.100.1.2/30 100.100.1.0 ether2
;;; PARA-R2
2 100.100.1.17/30 100.100.1.16 ether1
;;; PARA-R4
3 100.100.1.42/30 100.100.1.40 ether3
[admin@R6] >
```

Ao identificar o endereço da Rede configurado na interface da Loopback, o RouterOs criou uma rota diretamente conectada com a distância 0. Como por padrão a distância das rotas OSPF é 110, o router escolhia a rota com o menor custo e enviava pacotes para ele mesmo. O segundo efeito, era a falta de comunicação dos outros Routers com o R6 através do endereço de Loopback.

```
5 100.100.1.30/30 100.100.1.28 ether5
[admin@R2] > /ip/route/print where dst-address="10.100.0.2"
Flags: D - DYNAMIC; A - ACTIVE; c, o, y - COPY
Columns: DST-ADDRESS, GATEWAY, DISTANCE
DST-ADDRESS GATEWAY DISTANCE
Dc 10.100.0.2/32 loopback 0
[admin@R2] >

[admin@R6] > /ip/route/print where dst-address="10.100.0.2"
Flags: D - DYNAMIC; A - ACTIVE; c, o, y - COPY
Columns: DST-ADDRESS, GATEWAY, DISTANCE
D o 10.100.0.2/32 100.100.1.18 ether1 110
Dc 10.100.0.2/32 loopback 0
[admin@R6] >
```

9. Configurações adicionais no Firewall do R8

A fim de permitir apenas as conexões necessárias para a Lan da VPLS, foi configurado regras no firewall do R8 (Servidor DHCP) para impedir a comunicação com as Loopbacks e com os endereços das interfaces físicas dos Routers. Perceba que uma das regras possui uma exceção, permitindo o tráfego usando qualquer protocolo para o endereço do servidor de aplicação Nginx.

```
[admin@R8] > ip firewall/address-list/print
Columns: LIST, ADDRESS, CREATION-TIME
# LIST ADDRESS CREATION-TIME
0 IPS 10.100.0.0/24 aug/31/2026 01:48:22
1 IPS 100.100.1.0/24 aug/31/2026 02:14:10
[admin@R8] > ip firewall/filter/print
Flags: X - disabled, I - invalid; D - dynamic
0 ;;; Bloqueia VPC de acessar IPs dos Routers
chain=forward action=drop src-address=192.168.10.0/24
dst-address=!100.100.1.254 dst-address-list=IPS log=no log-prefix=""

1 ;;; Bloqueia VPC de acessar o R8 local
chain=input action=drop src-address=192.168.10.0/24
dst-address-list=IPS log=no log-prefix=""
```

10.Tabela de Endereçamento

A tabela mostra dois ranges livres, isso aconteceu devido há algumas alterações nos enlaces da infraestrutura do laboratório. Os endereços de Loopback seguem uma sequência lógica, indo de 1 a 13, atribuindo aos Routes respectivamente.

Sub-rede	Host 1	Host 2	Broadcast
100.100.1.0/30	100.100.1.1 (R1)	100.100.1.2 (R6)	100.100.1.3
100.100.1.4/30	100.100.1.5 (R11)	100.100.1.6 (R9)	100.100.1.7
100.100.1.8/30	100.100.1.9 (R5)	100.100.1.10 (R3)	100.100.1.11
100.100.1.12/30	100.100.1.13 (R3)	100.100.1.14 (R2)	100.100.1.15
100.100.1.16/30	100.100.1.17 (R6)	100.100.1.18 (R2)	100.100.1.19
100.100.1.20/30	100.100.1.21 (R2)	100.100.1.22 (R10)	100.100.1.23
100.100.1.24/30	100.100.1.25 (R2)	100.100.1.26 (R13)	100.100.1.27
100.100.1.28/30	100.100.1.29 (R1)	100.100.1.30 (R2)	100.100.1.31
100.100.1.32/30	100.100.1.33 (R1)	100.100.1.34 (R5)	100.100.1.35
100.100.1.36/30	100.100.1.37 (Livre)	100.100.1.38 (Livre)	100.100.1.39
100.100.1.40/30	100.100.1.41 (R4)	100.100.1.42 (R6)	100.100.1.43
100.100.1.44/30	100.100.1.45 (R4)	100.100.1.46 (R7)	100.100.1.47
100.100.1.48/30	100.100.1.49 (R7)	100.100.1.50 (R8)	100.100.1.51
100.100.1.52/30	100.100.1.53 (R8)	100.100.1.54 (R1)	100.100.1.55

Sub-rede	Host 1	Host 2	Broadcast
100.100.1.56/30	100.100.1.57 (Livre)	100.100.1.58 (Livre)	100.100.1.59
100.100.1.60/30	100.100.1.61 (R12)	100.100.1.62 (R1)	100.100.1.63
100.100.1.64/30	100.100.1.65 (R11)	100.100.1.66 (R1)	100.100.1.67
100.100.1.252/30	100.100.1.253 (R1)	100.100.1.254 (Nginx)	100.100.1.255

Range VPLS DHCP R1 R8 - - > 192.168.10.0/24			
192.168.10.0/24	192.168.10.1 (R8)	192.168.10.252 (VPC-R8)	192.168.10.255
192.168.10.0/24	192.168.10.1 (R8)	192.168.10.253 (VPC-R1)	192.168.10.255
192.168.10.0/24	192.168.10.1 (R8)	192.168.10.254 (R1)	192.168.10.255

